

## Functional & Performance Testing

### Model Performance Test

|               |                                        |
|---------------|----------------------------------------|
| Date          | 21 February 2026                       |
| Team ID       | LTVIP2026TMIDS38297                    |
| Project Name  | Gemini Historical Artifact Description |
| Maximum Marks |                                        |

| Test Case ID | Scenario (What to test)      | Test Steps (How to test)                                                  | Expected Result                                                        | Actual Result                                                      | Pass/Fail |
|--------------|------------------------------|---------------------------------------------------------------------------|------------------------------------------------------------------------|--------------------------------------------------------------------|-----------|
| FT-01        | Artifact Name Validation     | Enter valid names (e.g., "Pyramids") and empty strings in the text input. | Valid names accepted; empty input shows "Please enter a name" warning. | Warning displayed for empty input; valid name processes correctly. | Pass      |
| FT-02        | Word Count Slider Validation | Adjust the slider between 500 and 2000.                                   | Slider moves smoothly and updates the word_count variable.             | Slider updates value correctly.                                    | Pass      |
| FT-03        | Text Content Generation      | Enter "Viking Shield" and click "Generate Description".                   | Comprehensive Markdown description with 5 sections generated.          | Generated text includes all 5 requested sections.                  | Pass      |
| FT-04        | API Connection Check         | Check the sidebar status indicator on load.                               | Indicator shows "Gemini API Connected" (Green).                        | Sidebar shows "Connected".                                         | Pass      |
| FT-05        | Image Analysis Generation    | Upload a valid JPG image of a vase and click "Analyze".                   | System identifies object as "Greek Amphora" with details.              | Object correctly identified with details.                          | Pass      |
| PT-01        | Text Response Time           | Measure time from clicking "Generate" to result display.                  | Should be under 5-8 seconds (Gemini Flash model).                      | Average time was 4.2 seconds.                                      | Pass      |
| PT-02        | Image Upload Speed           | Upload a 5MB high-res image.                                              | Image preview should appear within 1 second.                           | Preview appeared in 0.8 seconds.                                   | Pass      |
| PT-03        | API Rate Limit Handling      | Click "Generate" 5 times in rapid succession.                             | System should handle requests or show "Too Many Requests" gracefully.  |                                                                    |           |